
GENERAL RELATIVITY

Problem set 1, 13.09.2019.

■ **PROBLEM 1** Problem 1.2 from Ref. [1]

Imagine that space (not spacetime) is actually a finite box, or in more sophisticated terms, a three-torus, of size L . By this, we mean that there is a coordinate system $x^\mu = (t, x, y, z)$ such that every point with coordinates (t, x, y, z) is *identified* with every point with coordinates $(t, x + L, y, z)$, $(t, x, y + L, z)$ and $(t, x, y, z + L)$. Note that the time coordinate is the same. Now consider two observers; observer A is at rest in this coordinate system (constant spatial coordinates), while observer B moves in the x -direction with constant velocity v . A and B begin at the same event, and while A remains still, B moves once around the universe and comes back to intersect the worldline of A without ever having to accelerate (since the universe is periodic). What are the relative proper times experienced in this interval by A and B ? Is this consistent with your understanding of Lorentz invariance?

■ **PROBLEM 2** Problem 1.4 from Ref. [1]

Projection effects can trick you into thinking that an astrophysical object is moving "superluminally". Consider a quasar that ejects gas with speed v at an angle θ with respect to the line-of sight with angular speed $\frac{v_{app}}{D}$, where D is the distance to the quasar and v_{app} is the apparent speed. Derive an expression for v_{app} in terms of v and θ . Show that, for appropriate values of v and θ , v_{app} can be greater than 1.

■ **PROBLEM 3** Problem 1.5 from Ref. [1]

Particle physicists are so used to setting $c = 1$ that they measure mass in units of energy. In particular, they tend to use electron volts ($1 \text{ eV} = 1.6 \times 10^{-12} \text{ erg} = 1.8 \times 10^{-33} \text{ g}$), or, more commonly, keV, MeV, and GeV (10^3 eV , 10^6 eV and 10^9 eV , respectively). The muon has been measured to have a mass of 0.106 GeV and a rest frame lifetime of 2.19×10^{-6} seconds. Imagine that such a muon is moving in the circular storage ring of a particle accelerator, 1 kilometer in diameter, such that the muon's total energy is 1000 GeV . How long would it appear to live from the experimenter's point of view? How many radians would it travel around the ring?

■ **PROBLEM 4** Lorentz covariant formulation of electromagnetism.

The Maxwell action for electromagnetism is

$$S = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{4\pi}{c} j^\mu A_\mu \right], \quad (4.1)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor, A_μ is the vector potential, and $j^\mu = (c\rho, \vec{j})$ is the electromagnetic (charge) current. Furthermore,

$$F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix}, \quad (4.2)$$

where $\eta^{\alpha\beta} = \eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1)$ is the Minkowski metric tensor, and $\vec{E} = (E^1, E^2, E^3)$ and $\vec{B} = (B^1, B^2, B^3)$ are the electric and magnetic fields, respectively.

- (a) By making use of the variational principle, $\delta S / \delta A_\mu(x) = 0$, show that the implied equation of motion is

$$\partial_\nu F^{\nu\mu} = \frac{4\pi}{c} j^\mu. \quad (4.3)$$

Show further that (4.3) implies the usual inhomogeneous Maxwell's equations,

$$\nabla \cdot \vec{E} = 4\pi\rho, \quad -\frac{1}{c} \frac{\partial}{\partial t} \vec{E} + \nabla \times \vec{B} = \frac{4\pi}{c} \vec{j}. \quad (4.4)$$

- (b) Show that the dual field tensor,

$$*F^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} = \begin{pmatrix} 0 & -B^1 & -B^2 & -B^3 \\ B^1 & 0 & E^3 & -E^2 \\ B^2 & -E^3 & 0 & E^1 \\ B^3 & E^2 & -E^1 & 0 \end{pmatrix}, \quad (4.5)$$

obeys a homogeneous equation,

$$\partial_\nu *F^{\nu\mu} = 0. \quad (4.6)$$

Here $\epsilon^{\mu\nu\alpha\beta}$ is the so-called Levi-Civita tensor, which is defined to be

$$\epsilon^{\mu\nu\alpha\beta} = \begin{cases} 1 & \text{if } (\mu\nu\alpha\beta) = \text{any even permutation of } (0123), \\ -1 & \text{if } (\mu\nu\alpha\beta) = \text{any odd permutation of } (0123), \\ 0 & \text{otherwise.} \end{cases} \quad (4.7)$$

Show further that these equations imply the homogeneous Maxwell's equations,

$$\nabla \cdot \vec{B} = 0, \quad \frac{1}{c} \frac{\partial}{\partial t} \vec{B} + \nabla \times \vec{E} = 0. \quad (4.8)$$

- (c) Recall that under Lorentz transformation $F^{\mu\nu}$ transforms as

$$F^{\mu\nu} \rightarrow F'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta F^{\alpha\beta}. \quad (4.9)$$

Consider first an ordinary rotation around the z -axis. Show that the magnetic and electric fields transform as ordinary vectors under such a rotation. Consider next a boost along the x -axis with velocity v . Show that we have the following transformation, mixing electric and magnetic fields ($\gamma = 1/\sqrt{1 - \beta^2}$, $\beta = v/c$),

$$\begin{aligned} B'_1 &= B_1, & B'_2 &= \gamma(B_2 + \beta E_3), & B'_3 &= \gamma(B_3 - \beta E_2), \\ E'_1 &= E_1, & E'_2 &= \gamma(E_2 - \beta B_3), & E'_3 &= \gamma(E_3 + \beta B_2). \end{aligned} \quad (4.10)$$

Recall that a Lorentz boost along the x -axis is given by the transformation

$$x' = \gamma(x - \beta x^0), \quad y' = y, \quad z' = z, \quad x'^0 = \gamma(x^0 - \beta x). \quad (4.11)$$

- (d) Consider now an inertial system $\mathbf{I}$ and the Lorentz boosted system $\mathbf{I}'$ (where $\mathbf{I}'$ is not rotated relative to $\mathbf{I}$), moving with velocity $\vec{v}$ relative to $\mathbf{I}$. The formula generalizing formula (4.10) above is

$$\vec{B} = \gamma \vec{B}' + \frac{\vec{v}}{v^2} (\vec{v} \cdot \vec{B}') (1 - \gamma) + \gamma \frac{\vec{v}}{c} \times \vec{E}' , \quad (4.12)$$

$$\vec{E} = \gamma \vec{E}' + \frac{\vec{v}}{v^2} (\vec{v} \cdot \vec{E}') (1 - \gamma) - \gamma \frac{\vec{v}}{c} \times \vec{B}' . \quad (4.13)$$

Assume that $\vec{E}'$ and $\vec{B}'$ are constant and different from zero. Find the condition that $\vec{E}'$ and $\vec{B}'$ have to satisfy in order that there exists a $\vec{v}$ such that $\vec{E} = 0$, and find the corresponding $\vec{v}$ expressed in terms of $\vec{E}'$ and $\vec{B}'$. (Remark: by considering charged particles in this special Lorentz frame one easily finds how they will move.)

- (e) Show that the two quantities $\vec{E} \cdot \vec{B}$ and $|\vec{E}|^2 - |\vec{B}|^2$ are lorentz invariant, *i.e.* they are Lorentz scalars.

Hint: You might find the following identities useful,

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) , \quad (4.14)$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C} , \quad (4.15)$$

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D}) . \quad (4.16)$$

Try to prove these identities by using the antisymmetric symbol, ϵ^{ijk} . Recall that $(\vec{A} \times \vec{B})^i = \epsilon^{ijk} A^j B^k$.

References

- [1] Sean M. Carroll, “An introduction to General Relativity, Spacetime and Geometry” Addison Wesley (2004), ISBN 0-8053-8732-3.